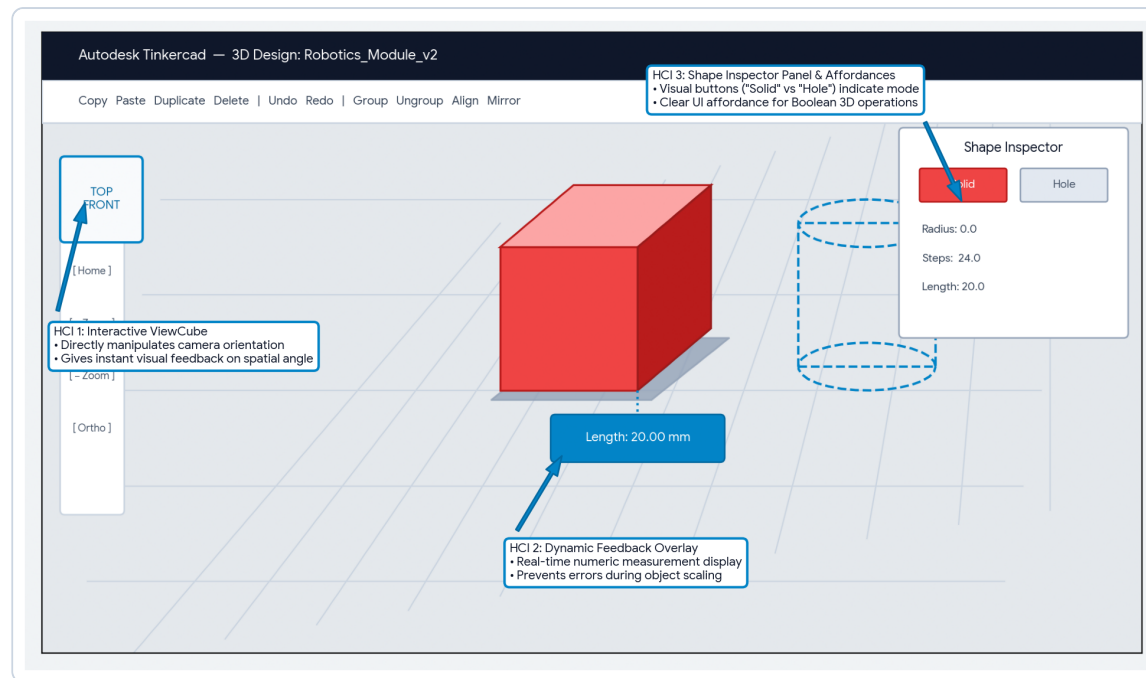


HCI Breakdown: Autodesk Tinkercad (Interactive 3D Tool)

Slide 1 of 2 — HCI Focus



INTERACTION & UI ELEMENTS (HCI)

1. ViewCube Navigation Widget

Usability Purpose: Provides direct manipulation of the 3D camera orientation. Allows users to switch seamlessly between predefined orthographic standard views (Top, Front, Side) or smoothly rotate the viewport without memorizing complex key bindings.

2. Dynamic Feedback Overlay

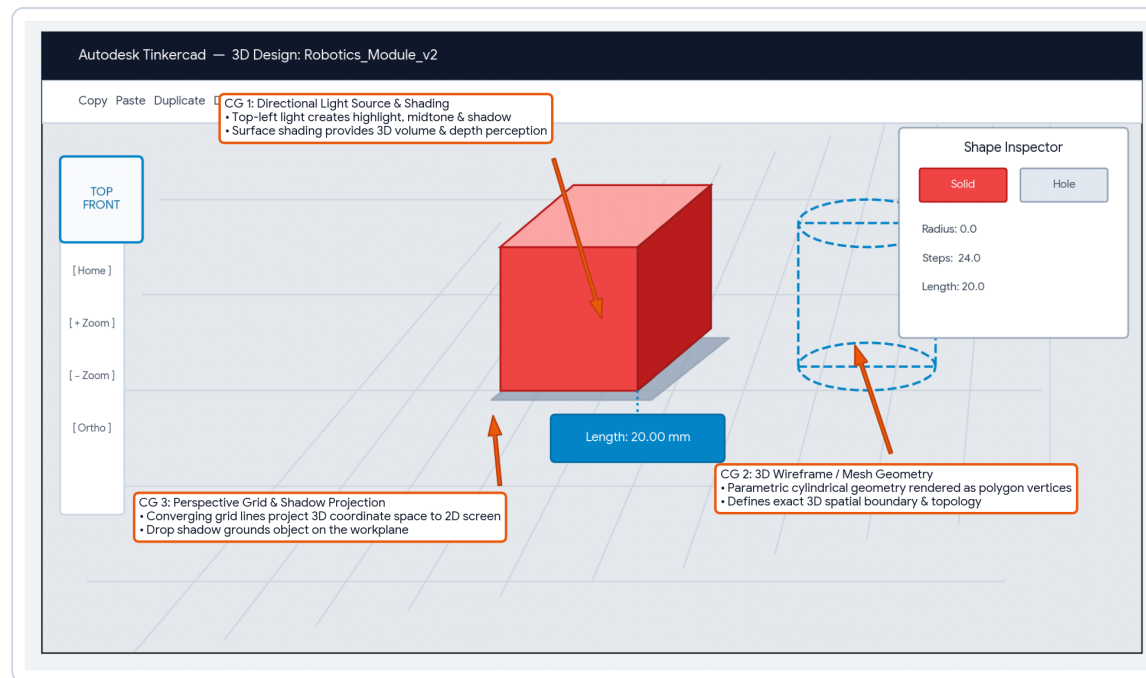
Usability Purpose: Offers real-time numerical feedback showing exact object dimensions during scaling operations. Ensures precision modeling, minimizes cognitive load, and eliminates user guesswork.

3. Visual Affordance Inspector Panel

Usability Purpose: Uses clear visual toggles ("Solid" vs. "Hole") to indicate geometric subtraction states. Gives immediate visual affordance on Boolean operations before execution.

Computer Graphics Breakdown & UX Reflection

Slide 2 of 2 — CG Focus & Reflection



RENDERING & GRAPHICS ELEMENTS (CG)

1. Directional Light & Shading

Rendering Function: Simulates a fixed light source producing diffuse shading (highlights, midtones, shadows) across polygon faces to convey 3D form and volume on a 2D screen.

2. Polygon Mesh Geometry

Rendering Function: Defines 3D objects via vertex vectors and connected wireframe polygons (lines and faces), providing the fundamental mathematical representation of 3D spatial boundaries.

3. Perspective Projection & Ground Shadow

Rendering Function: Utilizes perspective transformation matrices to converge parallel lines into the distance, combined with drop shadow rendering to ground models firmly on the workplane coordinate system.

Q1: How does CG quality help or hinder the user's ability to interact with the software?

Answer: Computer Graphics quality directly aids usability by providing vital spatial visual cues, such as directional shading and drop shadows, which help users accurately gauge object depth and alignment on a flat screen. High-quality real-time rendering prevents perceptual misinterpretations during 3D placement. However, overly complex realistic lighting or dense photorealistic shaders can hinder interaction by causing visual clutter, obscuring fine geometric edges, and inducing GPU performance lag during real-time editing.

Q2: What is one simple HCI change you would make to improve the user interface?

Answer: I would add dynamic smart alignment snap lines with contextual distance pop-ups when moving objects close to one another (similar to Figma's smart guides, but in 3D space). Currently, aligning objects requires opening a separate modal tool or manual camera rotation to check all three axes. Implementing auto-sensing edge snap guides would significantly streamline spatial layout tasks and reduce user navigation friction.